

Lecture 12

Invariant integration, characters, and Wigner–Eckart

Objectives. *Part I ran on a single engine: the average over the group. This lecture rebuilds that engine for compact groups — the invariant integral in place of the sum, constructed explicitly for $U(1)$ and $SU(2)$, with the class integration formula $\frac{1}{\pi} \int f(\theta) \sin^2(\theta/2) d\theta$ proved — and collects the dividends at once: every representation is unitarisable (closing Lecture 10's deferred clause), characters are orthonormal and complete, and the character formula $\chi_j = \sin((2j+1)\theta/2)/\sin(\theta/2)$ repays, with interest, the loan Lecture 7 took for the crystal field. Tensor products then answer the question experiment asks daily — what happens when two angular momenta meet — through the Clebsch–Gordan series and coefficients; the Wigner–Eckart theorem compresses whole families of matrix elements into single numbers; and the applications cash out as dipole selection rules and the Landé g -factor.*

12.1 Invariant integration

Everything Part I proved flowed from one device: the average $\frac{1}{|G|} \sum_{g \in G}$, whose two virtues were that it exists and that translating the summation variable costs nothing. For a continuous group the sum is meaningless, but the virtues survive in integral form. The general statement is our next-to-last conceptual import.

The representation category in this lecture is continuous and finite-dimensional unless an L^2 function space is named explicitly. Accordingly the averaging, Schur, character, and tensor-product theorems below state finite-dimensional hypotheses. They are compact-group analogues, not a licence to apply finite-group dimension counts to arbitrary Hilbert-space representations.

Theorem 12.1 (Haar measure; imported). *Every compact matrix Lie group G carries a unique Borel probability measure dg (Borel: defined on every set one would integrate over) that is left-invariant: $\int_G f(hg) dg = \int_G f(g) dg$ for all $h \in G$ and continuous f . This measure is automatically right-invariant and invariant under $g \mapsto g^{-1}$.*

Reference for the omitted proof: Hall, Ch. 4, Theorem 4.28, and Appendix D on Haar measure.

We import the theorem for culture and honesty, but we will never integrate over a group whose measure we have not built by hand. For $U(1)$ the measure is $d\theta/2\pi$ on $[0, 2\pi)$, and invariance is the translation invariance of Lebesgue measure plus periodicity — two lines, both the reader's. For $SU(2)$ the construction is geometry.

Proposition 12.2 (The Haar measure of SU(2)). Identify SU(2) with S^3 by $U(x) = x_0\mathbb{1} - i\vec{x} \cdot \vec{\sigma}$, $x \in S^3 \subset \mathbb{R}^4$ (Lecture 10’s quaternion picture), and for continuous f set

$$\int_{\text{SU}(2)} f(U) dU := \frac{1}{2\pi^2} \int_{S^3} f(U(x)) dS(x),$$

the normalised surface integral ($\text{vol } S^3 = 2\pi^2$). This is a probability measure invariant under left translation, right translation, and inversion.

Proof. The coordinates obey $\sum_{\mu} x_{\mu}^2 = \frac{1}{2} \text{tr}(U(x)^{\dagger}U(x))$, a direct Pauli computation. Fix $V \in \text{SU}(2)$. The quaternion subspace $\text{span}_{\mathbb{R}}\{\mathbb{1}, -i\sigma_a\}$ is closed under matrix products (Lecture 10), so $w \mapsto Vw$ maps it to itself; the map is linear in the coordinates x , and it preserves the quadratic form above because $\text{tr}((Vw)^{\dagger}(Vw)) = \text{tr}(w^{\dagger}V^{\dagger}Vw) = \text{tr}(w^{\dagger}w)$. Left translation is therefore an *orthogonal* linear map of $\mathbb{R}^4$; it preserves S^3 and its surface measure (orthogonal maps preserve Lebesgue measure on $\mathbb{R}^4$, and the surface measure on S^3 is defined as the limit of the volumes of thin shells, which they therefore preserve too). The same argument word for word handles right translation $w \mapsto wV$, and inversion is $U \mapsto U^{\dagger}$, i.e. $x \mapsto (x_0, -\vec{x})$, orthogonal again. ■

The first dividend settles Lecture 10’s largest outstanding debt.

Theorem 12.3 (Unitarisability, and Maschke redux). Let G be a compact matrix Lie group with an invariant integral (for us: U(1), SU(2), and SO(3) via Sheet 12). Then:

- (i) every finite-dimensional continuous representation Π of G admits an invariant inner product, namely

$$\langle v, w \rangle_G = \int_G \langle \Pi(g)v, \Pi(g)w \rangle dg,$$

and is therefore equivalent to a unitary representation;

- (ii) every finite-dimensional continuous representation of G is a direct sum of irreducible subrepresentations.

In particular clause (ii) of Lecture 10’s complete reducibility theorem — stated there, deferred here — is now proved.

Proof. (i) Sesquilinearity is inherited from the original inner product and the linearity of the integral. The integrand is continuous on a compact group, so the integral exists. If $v \neq 0$, then $\Pi(g)v \neq 0$ for every g , since $\Pi(g)$ is invertible with inverse $\Pi(g^{-1})$. Thus $g \mapsto \|\Pi(g)v\|^2$ is continuous and strictly positive on compact G , so it has a positive minimum m ; because the invariant integral is normalised, $\langle v, v \rangle_G \geq m > 0$. Finally, for $h \in G$, $\langle \Pi(h)v, \Pi(h)w \rangle_G = \int_G \langle \Pi(gh)v, \Pi(gh)w \rangle dg = \langle v, w \rangle_G$ by right invariance — Lecture 3’s averaging proof with the sum retired. (ii) Combine (i) with the orthocomplement induction already proved in Lecture 10. ■

So the Gaussian inner product of Lecture 10 was not a fluke but an instance: averaging now manufactures an invariant inner product on *any* representation, and on an irreducible one the two constructions agree up to scale (Schur: writing $\langle v, w \rangle_2 = \langle Av, w \rangle_1$ defines an operator A commuting with the representation, hence a scalar).

For characters one more tool is needed: integration over conjugacy classes. Here SU(2) is unusually kind.

Proposition 12.4 (The classes of $SU(2)$). Every $U \in SU(2)$ is conjugate, within $SU(2)$, to exactly one

$$u_\theta = e^{-i\theta\sigma_3/2} = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}, \quad \theta \in [0, 2\pi],$$

so conjugacy classes are labelled by $\text{tr} U = 2 \cos(\theta/2) \in [-2, 2]$. Under $\Phi(u_\theta) = R_z(\theta)$, the parameter θ is the geometric rotation angle.

Proof. The finite-dimensional spectral theorem (declared linear-algebra prerequisite, not the unbounded functional-calculus import of Lectures 9 and 13) diagonalises the unitary U by some $W \in U(2)$; multiplying W by a phase puts it in $SU(2)$ without changing the conjugation. With $\det U = 1$ the eigenvalues are a pair $e^{\pm i\theta/2}$, and conjugating by $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in SU(2)$ swaps them (replacing θ by $-\theta$; and θ is only defined modulo 4π), so θ may be taken in $[0, 2\pi]$, where $\cos(\theta/2)$ is injective: the label is unique, and it is the trace. ■

In the sphere picture the classes become visible: $\text{tr} U(x) = 2x_0$, so the conjugacy classes of $SU(2)$ are the latitudes of S^3 — the two poles $\pm\mathbb{K}$ are the one-point classes $\theta = 0, 2\pi$, and the equator $\theta = \pi$ (the traceless unitaries) is the fattest class. Integrating a class function is then a matter of weighing latitudes by their area.

Theorem 12.5 (Class integration formula). For every continuous class function f on $SU(2)$,

$$\int_{SU(2)} f(U) \, dU = \frac{1}{\pi} \int_0^{2\pi} f(u_\theta) \sin^2 \frac{\theta}{2} \, d\theta. \quad (12.1)$$

Proof. In hyperspherical coordinates $x_0 = \cos \psi$, $\vec{x} = \sin \psi \, \hat{n}$ with $\psi \in [0, \pi]$, $\hat{n} \in S^2$, the surface element is $dS = \sin^2 \psi \, d\psi \, d\Omega$ — the four-dimensional polar Jacobian from the analysis course; the check $\int_0^\pi \sin^2 \psi \, d\psi \cdot 4\pi = 2\pi^2$ recovers the volume. A class function depends only on the latitude, $f(U(x)) = f(u_{2\psi})$ by Proposition 12.4, so the angular integral contributes its full 4π and

$$\int_{SU(2)} f \, dU = \frac{4\pi}{2\pi^2} \int_0^\pi f(u_{2\psi}) \sin^2 \psi \, d\psi = \frac{1}{\pi} \int_0^{2\pi} f(u_\theta) \sin^2 \frac{\theta}{2} \, d\theta$$

after $\theta = 2\psi$. (Normalisation check: $\frac{1}{\pi} \int_0^{2\pi} \sin^2(\theta/2) \, d\theta = 1$.) ■

The weight $\sin^2(\theta/2)$ is therefore not a formula to memorise but a picture: the relative area of the latitude sphere of radius $\sin(\theta/2)$. Finally, $SO(3)$ inherits its Haar measure through the double cover — a function on $SO(3)$ pulls back along Φ to an even function on $SU(2)$, and one defines $\int_{SO(3)} f := \int_{SU(2)} f \circ \Phi$, invariance included; Sheet 12 supplies the bookkeeping.

12.2 Characters, again

The definitions of Part I are group-agnostic and port verbatim: the *character* of a finite-dimensional representation is $\chi_\Pi(g) = \text{tr} \Pi(g)$; it is continuous, constant on conjugacy classes, invariant under equivalence, additive over direct sums, and satisfies $\chi_\Pi(e) = \dim \Pi$ and, for unitary Π — which Theorem 12.3 lets us assume throughout — $\chi_\Pi(g^{-1}) = \overline{\chi_\Pi(g)}$. What Part I could not give us without an average is the orthogonality; with the integral it can.

Theorem 12.6 (Orthogonality of characters). *Let Π, Σ be irreducible unitary representations of a compact group G with invariant integral. Then*

$$\langle \chi_\Pi, \chi_\Sigma \rangle := \int_G \overline{\chi_\Pi(g)} \chi_\Sigma(g) dg = \begin{cases} 1 & \Pi \cong \Sigma, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. Lecture 4’s proof, with the sum retired. For a linear map X from Σ ’s space to Π ’s, average the conjugated map: $T_X = \int_G \Pi(g) X \Sigma(g)^{-1} dg$. Left invariance gives $\Pi(h) T_X \Sigma(h)^{-1} = T_X$: an intertwiner. If $\Pi \not\cong \Sigma$, Schur’s lemma (Lecture 3) forces $T_X = 0$ for every X ; choosing X to run over matrix units and using $\Sigma(g)^{-1} = \Sigma(g)^\dagger$, the entries read

$$\int_G \Pi(g)_{il} \overline{\Sigma(g)_{jk}} dg = 0,$$

and summing over $i = l, j = k$ kills $\langle \chi_\Sigma, \chi_\Pi \rangle$. If $\Pi = \Sigma$ (dimension d), Schur gives $T_X = c(X) \mathbb{1}$ with $c(X) = \text{tr}(T_X)/d = \text{tr}(X)/d$, since conjugation preserves traces under the integral; the same entry-reading now yields $\int \Pi_{il} \overline{\Pi_{jk}} dg = \delta_{ij} \delta_{lk}/d$, and the double trace gives $\langle \chi_\Pi, \chi_\Pi \rangle = d/d = 1$. ■

Corollary 12.7 (The character calculus). *For a compact group with invariant integral, with complete reducibility now proved: the multiplicity of an irreducible Σ in Π is $\langle \chi_\Sigma, \chi_\Pi \rangle$; a representation is irreducible if and only if $\langle \chi_\Pi, \chi_\Pi \rangle = 1$; two representations with equal characters are equivalent; and the projection onto the Σ -isotypic component is $P_\Sigma = \dim \Sigma \int_G \overline{\chi_\Sigma(g)} \Pi(g) dg$. The proofs are Part I’s, verbatim.*

For $U(1)$ the irreducible characters are $\chi_n(e^{i\theta}) = e^{in\theta}$, $n \in \mathbb{Z}$ (Lecture 9), and orthogonality reads $\frac{1}{2\pi} \int_0^{2\pi} e^{i(n-m)\theta} d\theta = \delta_{nm}$: *Fourier orthogonality*. Completeness of the exponentials in L^2 — a point-of-use import, like the sine-series completeness used below, not an assumed prerequisite — says the characters of $U(1)$ miss nothing. Fourier series, in other words, is the character theory of the circle, a sentence that closes the thread Lecture 4 opened when finite characters first behaved like frequencies.

Theorem 12.8 (The characters of $SU(2)$). *On the class of u_θ ,*

$$\chi_j(\theta) = \sum_{m=-j}^j e^{-im\theta} = \frac{\sin((2j+1)\theta/2)}{\sin(\theta/2)}, \quad \theta \in (0, 2\pi), \quad (12.2)$$

with the limiting values $\chi_j(0) = 2j+1$ and $\chi_j(2\pi) = (-1)^{2j}(2j+1)$.

Proof. $\Pi_j(u_\theta) = e^{-i\theta J_z}$ has eigenvalues $e^{-im\theta}$, $m = -j, \dots, j$, by the classification theorem; the trace is a geometric sum:

$$\sum_{m=-j}^j e^{-im\theta} = \frac{e^{ij\theta} - e^{-i(j+1)\theta}}{1 - e^{-i\theta}} = \frac{e^{i(j+\frac{1}{2})\theta} - e^{-i(j+\frac{1}{2})\theta}}{e^{i\theta/2} - e^{-i\theta/2}},$$

after multiplying through by $e^{i\theta/2}$. The endpoint values are the sums directly — and $\chi_j(2\pi) = (-1)^{2j}(2j+1)$ is $\text{tr} \Pi_j(-\mathbb{1})$, Lecture 10’s sign flip taking attendance. ■

Here, then, is the repayment Part I has been waiting for since the crystal field. Lecture 7 split the atomic d -shell in an octahedral environment by borrowing, on credit, the character of a rotation by θ acting on the $2\ell+1$ orbital states: $\sin((\ell + \frac{1}{2})\theta)/\sin(\theta/2)$. For integer

$j = \ell$ the representation and its character descend to $\mathrm{SO}(3)$ (Lecture 11), θ is the geometric rotation angle (Proposition 12.4), and (12.2) is exactly the borrowed function: every step of that crystal-field computation is now a theorem. Lecture 11 promised that this lecture would need only the *traces* of the Wigner matrices; (12.2) is those traces, earning their keep.

Orthonormality can be checked against the class measure by hand: by (12.1), with $\psi = \theta/2$,

$$\langle \chi_j, \chi_{j'} \rangle = \frac{2}{\pi} \int_0^\pi \sin((2j+1)\psi) \sin((2j'+1)\psi) d\psi = \delta_{jj'},$$

the orthogonality of sines on $[0, \pi]$. The measure's $\sin^2(\theta/2)$ cancels the characters' denominators: the Weyl weight and the characters were made for each other. Better still, the same change of variables proves completeness, with no machinery beyond second-year Fourier series.

Theorem 12.9 (Completeness of the $\mathrm{SU}(2)$ characters). *Let $L^2_{\mathrm{cl}}(\mathrm{SU}(2))$ be the Hilbert space of square-integrable class functions, where functions equal almost everywhere are identified. The characters $\{\chi_j : j \in \frac{1}{2}\mathbb{Z}_{\geq 0}\}$ are an orthonormal basis of this space. In particular, every square-integrable class function is the L^2 -limit of finite linear combinations of characters.*

Proof. The class parameter is $\theta \in [0, 2\pi]$, with the two endpoints representing the distinct one-point classes $\mathbb{K}$ and $-\mathbb{K}$; both have zero Weyl weight and hence do not affect L^2 classes. The integration formula (12.1) extends from continuous to integrable class functions by approximating them with continuous ones (density of continuous functions, a declared prerequisite). With $\theta = 2\psi$, it therefore makes

$$Tf(\psi) = \sqrt{\frac{2}{\pi}} f(u_{2\psi}) \sin \psi, \quad 0 < \psi < \pi,$$

an isometry from $L^2_{\mathrm{cl}}(\mathrm{SU}(2))$ to $L^2(0, \pi)$:

$$\|f\|_{L^2(\mathrm{SU}(2))}^2 = \frac{2}{\pi} \int_0^\pi |f(u_{2\psi})|^2 \sin^2 \psi d\psi = \|Tf\|_{L^2(0, \pi)}^2.$$

It is onto: for $h \in L^2(0, \pi)$ prescribe the class function by $f(u_{2\psi}) = \sqrt{\pi/2} h(\psi)/\sin \psi$ for $0 < \psi < \pi$, assigning arbitrary endpoint values. Moreover,

$$T\chi_j(\psi) = \sqrt{\frac{2}{\pi}} \sin((2j+1)\psi).$$

We now import at this point the Fourier sine completeness theorem (Stein–Shakarchi, *Fourier Analysis*, Ch. 1, §1, and Ch. 3, Theorem 1.1): $\{\sqrt{2/\pi} \sin(n\psi) : n = 1, 2, \dots\}$ is an orthonormal basis of $L^2(0, \pi)$. As $j = 0, \frac{1}{2}, 1, \dots$, the integer $2j+1$ runs through every positive integer. Hence the characters map to that entire basis, proving the claim for arbitrary L^2 class functions, not only continuous ones. ■

Remark 12.10 (Peter–Weyl, starred). What we have proved by hand for $\mathrm{U}(1)$ and $\mathrm{SU}(2)$ holds for every compact group: the matrix elements of the irreducible representations are dense in $L^2(G)$, and the irreducible characters are complete among class functions — the Peter–Weyl theorem, the proper generalisation of both Fourier analysis and the finite counting and group-algebra statements separated in Lecture 7 from their compact analogues. We state it for perspective and never use more of it than we have proved.

12.3 Tensor products and Clebsch–Gordan

Quantum systems compose by tensor product: two subsystems with spaces V and W form one system with space $V \otimes W$, and a symmetry acts on both parts at once,

$$(\Pi_1 \otimes \Pi_2)(g) = \Pi_1(g) \otimes \Pi_2(g), \quad \chi_{\Pi_1 \otimes \Pi_2} = \chi_{\Pi_1} \chi_{\Pi_2}$$

(the trace of a Kronecker product is the product of traces — Lecture 7 multiplied characters this way already, and the construction is unchanged). Lecture 10's $\mathcal{H} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^{2s+1}$ was an instance. What is genuinely new is the derivative.

Proposition 12.11 (The product rule). *The Lie algebra representation derived from $\Pi_1 \otimes \Pi_2$ is*

$$\pi(X) = \pi_1(X) \otimes \mathbb{I} + \mathbb{I} \otimes \pi_2(X). \quad (12.3)$$

Proof. Write $A(t) = \Pi_1(e^{tX})$, $B(t) = \Pi_2(e^{tX})$ and expand the difference quotient bilinearly:

$$A(t) \otimes B(t) - A(0) \otimes B(0) = (A(t) - A(0)) \otimes B(t) + A(0) \otimes (B(t) - B(0));$$

divide by t and let $t \rightarrow 0$. It is the product rule, honestly. ■

In physics notation: the total angular momentum of a composite system is $\vec{J} = \vec{J}_1 \otimes \mathbb{I} + \mathbb{I} \otimes \vec{J}_2$, written $\vec{J}_1 + \vec{J}_2$ by everyone — additivity of angular momentum is the product rule in Hermitian dress. The question of the section is how the composite multiplets decompose, and the characters answer it in four lines.

Theorem 12.12 (The Clebsch–Gordan series). *For all $j_1, j_2 \in \frac{1}{2}\mathbb{Z}_{\geq 0}$,*

$$\Pi_{j_1} \otimes \Pi_{j_2} \cong \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} \Pi_j, \quad (12.4)$$

the sum in integer steps, each summand occurring exactly once.

Proof. Take $j_1 \geq j_2$. On the class of u_θ , use the sum form of χ_{j_2} and the closed form of χ_{j_1} :

$$\chi_{j_1} \chi_{j_2} = \sum_{m=-j_2}^{j_2} \frac{e^{-im\theta} (e^{i(j_1+\frac{1}{2})\theta} - e^{-i(j_1+\frac{1}{2})\theta})}{e^{i\theta/2} - e^{-i\theta/2}}.$$

In the second half of the numerator substitute $m \mapsto -m$ (the sum is over a symmetric range); the terms regroup into

$$\chi_{j_1} \chi_{j_2} = \sum_{m=-j_2}^{j_2} \frac{\sin((j_1 + \frac{1}{2} - m)\theta)}{\sin(\theta/2)} = \sum_{j=j_1-j_2}^{j_1+j_2} \chi_j(\theta),$$

since $j_1 + \frac{1}{2} - m$ runs over $j + \frac{1}{2}$ for $j = j_1 - j_2, \dots, j_1 + j_2$. Equal characters mean equivalent representations (Corollary 12.7), and the series is read off; multiplicity one because each χ_j appears once. As a check, the dimensions telescope: $\sum_{j=|j_1-j_2|}^{j_1+j_2} (2j+1) = (2j_1+1)(2j_2+1)$. Sheet 12 gives the second proof, by counting J_z -weights — same answer, different muscle. ■

The series tells us *which* multiplets appear; experiments need the change of basis. The product basis $|j_1, m_1\rangle \otimes |j_2, m_2\rangle$ and the coupled basis $|j, m\rangle$, $j = |j_1 - j_2|, \dots, j_1 + j_2$, are both orthonormal (multiplets of different j are orthogonal, being eigenspaces of the Hermitian

Casimir for distinct eigenvalues, as in Lecture 11), and the unitary matrix between them defines the *Clebsch–Gordan coefficients*,

$$|j, m\rangle = \sum_{m_1+m_2=m} \langle j_1 m_1; j_2 m_2 | j m \rangle |j_1, m_1\rangle \otimes |j_2, m_2\rangle,$$

fixed up to phases. Lecture 10’s positive ladder coefficients have already fixed the phases *within* each factor multiplet. The coupled-basis Condon–Shortley convention used here fixes the remaining relative phases as follows, with the tensor factors kept in the printed order. Construct the coupled multiplets in descending j . At each new top weight $m = j$, take the normalised vector annihilated by the total J_+ and orthogonal to the higher- j multiplets already constructed, and choose its overall sign so that the coefficient of the product ket with the largest allowed m_1 is positive and real. Then define the rest of that entire multiplet by Lecture 10’s positive total- J_- ladder formula. Multiplicity one in (12.4) makes each step unique.

Thus the top state $|j_1+j_2, j_1+j_2\rangle = |j_1, j_1\rangle \otimes |j_2, j_2\rangle$ is forced and positive; lower it to build that multiplet, take the orthogonal highest-weight direction at the next j , apply the stated sign rule, and repeat. Reversing the sign of *every* ket in one coupled multiplet gives an equivalent coupled basis and changes no representation-theoretic or physical result. Such a consistently sign-reversed answer is accepted unless a question explicitly requests the adopted Condon–Shortley phase; relative sign changes within a multiplet are not equivalent. The smallest case deserves to be done in public.

Before the algebra, the answer in one picture (Figure 12.1): laid out by weight $m = m_1+m_2$, the four states of $\frac{1}{2} \otimes \frac{1}{2}$ regroup into a triplet and a singlet.

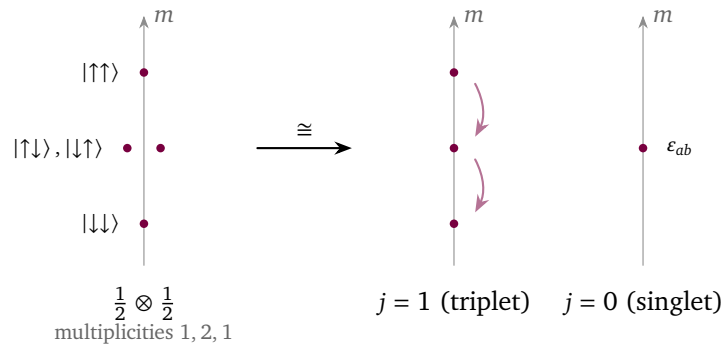


Figure 12.1: Clebsch–Gordan decomposition $\frac{1}{2} \otimes \frac{1}{2} \cong 1 \oplus 0$, read off the weights. The four product states distribute over $m = m_1 + m_2$ with multiplicities 1, 2, 1 (left). The two-fold $m = 0$ level resolves into the symmetric combination, which completes the triplet $j = 1$, and the antisymmetric combination $|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle$, the singlet $j = 0$ — the invariant tensor ε_{ab} of $SU(2)$. This is the smallest instance of the general rule $j_1 \otimes j_2 = \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} j$, and the source of the ε -tensor promised in Part I.

Example 12.13 (Two spin-halves: singlet and triplet). Let $j_1 = j_2 = \frac{1}{2}$, with product basis $|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle$. The series (12.4) says $\frac{1}{2} \otimes \frac{1}{2} \cong 0 \oplus 1$. Top state: $|1, 1\rangle = |\uparrow\uparrow\rangle$. Lower it: since $J_- |\uparrow\rangle = |\downarrow\rangle$, the product rule gives $J_- |\uparrow\uparrow\rangle = |\downarrow\uparrow\rangle + |\uparrow\downarrow\rangle$, while the ladder formula demands $J_- |1, 1\rangle = \sqrt{2} |1, 0\rangle$; hence

$$|1, 0\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle), \quad |1, -1\rangle = |\downarrow\downarrow\rangle \quad (\text{one more rung}),$$

and the lone direction orthogonal to $|1, 0\rangle$ in the $m = 0$ plane is the singlet up to sign. Here the largest allowed first-factor weight is $m_1 = \frac{1}{2}$, belonging to $|\uparrow\downarrow\rangle$, so the coupled-basis convention above selects the *singlet*

$$|0, 0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle), \quad J_{\pm} |0, 0\rangle = 0 \quad (\text{check it: the two terms cancel}).$$

Every Clebsch–Gordan coefficient of the problem is on display. The physics is half of low-temperature science: two spin- $\frac{1}{2}$ particles carry total spin 1 (the symmetric triplet) or 0 (the antisymmetric singlet) — the decomposition is precisely the symmetric/antisymmetric split of $\mathbb{C}^2 \otimes \mathbb{C}^2$ under the exchange action of S_2 , Part I’s smallest group commuting with $SU(2)$ ’s biggest job, a duet that Lecture 13 will turn into a chorus. The singlet is the maximally entangled Bell state; the singlet–triplet gap organises helium’s spectrum and the hydrogen molecule’s binding.

The singlet also settles a promise about *invariant tensors*. In components, $|0, 0\rangle = \frac{1}{\sqrt{2}} \sum_{a,b} \epsilon_{ab} e_a \otimes e_b$ with $\epsilon_{12} = -\epsilon_{21} = 1$, and its invariance is the cofactor identity: for any 2×2 matrix, $A^T \epsilon A = (\det A) \epsilon$ — expand it; it is the definition of the determinant — so ϵ is invariant under exactly the matrices of determinant one, $SU(2)$ included. The ϵ -tensor is the singlet wearing indices, and together with the Hermitian pairing $\delta_{a\bar{b}}$ it generates all invariant tensors of $SU(2)$ (stated). The analogous $SU(3)$ classification is outside this course: the dual/conjugate language of Lecture 3 and the invariant-tensor language of this paragraph are preparation for it, while Jones, Ch. 8, §§8.3–8.4, is the explicit further-reading destination.

12.4 Tensor operators and the Wigner–Eckart theorem

States organise into multiplets; the last step is to organise the *operators* — because what experiment measures is matrix elements $\langle \text{final} | T | \text{initial} \rangle$ of perturbations and couplings, and these inherit the symmetry too.

Definition 12.14 (Tensor operator). A tensor operator of rank k on a unitary representation $(\Pi, \mathcal{H})$ of $SU(2)$ is a family T_q^k , $q = -k, \dots, k$, of bounded, everywhere-defined operators $T_q^k \in \mathcal{B}(\mathcal{H})$ transforming among themselves like the spin- k multiplet:

$$\Pi(U) T_q^k \Pi(U)^{-1} = \sum_{q'} D_{q'q}^k(U) T_{q'}^k. \quad (12.5)$$

Proposition 12.15 (Group form = algebra form). On a finite-dimensional $\mathcal{H}$, (12.5) is equivalent to the bracket relations

$$[J_z, T_q^k] = q T_q^k, \quad [J_{\pm}, T_q^k] = \sqrt{k(k+1) - q(q \pm 1)} T_{q \pm 1}^k.$$

Proof. Conjugation $T \mapsto \Pi(U) T \Pi(U)^{-1}$ is itself a continuous representation of $SU(2)$ on the operator space $\mathfrak{gl}(\mathcal{H})$, with derivative $T \mapsto [\pi(X), T]$ (differentiate, as for Ad in Lecture 9). Differentiating (12.5) therefore gives $[\pi(X), T_q^k] = \sum_{q'} \pi_k(X)_{q'q} T_{q'}^k$, and inserting the ladder basis with Lecture 10’s matrix elements yields exactly the stated brackets. Conversely, the brackets say that the span of the T_q^k is an invariant subspace of $\mathfrak{gl}(\mathcal{H})$ on which the derived representation acts by the spin- k matrices; since $SU(2)$ is connected, Lecture 9’s dictionary integrates this back to (12.5). ■

The bounded hypothesis is the literal scope of the definition and theorem below. There is one standard domain-qualified extension used in physics. Let $\mathcal{D} \subset \mathcal{H}$ be a common dense $\text{SU}(2)$ -invariant domain, so $\Pi(U)\mathcal{D} = \mathcal{D}$, preserved by every T_q^k and by the derived generators. Then (12.5) may instead be read as an identity on $\mathcal{D}$: for $v \in \mathcal{D}$ the conjugated product means $\Pi(U)T_q^k\Pi(U)^{-1}v$, and all algebra-form commutators are identities on that same $\mathcal{D}$. Wigner–Eckart then applies to matrix elements between finite spin multiplets contained in $\mathcal{D}$; its proof uses only finite-dimensional restrictions and compressions. It does *not* make an unbounded operator everywhere defined.

Rank 0 includes a bounded operator commuting with all rotations (for example a bounded Borel function of a rotationally invariant self-adjoint Hamiltonian). Rank 1 is a *vector operator* $\vec{V}$ repackaged in spherical components

$$V_0 = V_z, \quad V_{\pm 1} = \mp \frac{1}{\sqrt{2}} (V_x \pm iV_y),$$

which is the harmonic basis of Lecture 11’s spin-one example, reappearing on schedule. Position, momentum, orbital angular momentum, and magnetic moment are generally *unbounded* vector-operator examples, outside the literal bounded definition: on $L^2(\mathbb{R}^3)$ one may take $\mathcal{D} = \mathcal{S}(\mathbb{R}^3)$, the Schwartz space of smooth functions all of whose derivatives decay faster than any power, which rotations and the operators $Q_i = x_i$, $P_i = -i\partial_i$, L_i preserve, and read both group and commutator formulas there. Spin matrices on a fixed finite spin space are, of course, bounded. By contrast each fixed spherical harmonic $Y_{\ell m}$ is continuous and bounded on the compact sphere, so multiplication by it on $L^2(S^2)$ is genuinely a bounded tensor operator of rank ℓ ; conjugation by rotations moves its argument, which is (12.5) for the function action.

Theorem 12.16 (Wigner–Eckart). *Let T^k be a bounded tensor operator in Definition 12.14. Let $|\alpha j m\rangle$ be a ladder-normalised basis (Lecture 10’s ladder phases) of an invariant spin- j multiplet and $|\alpha' j' m'\rangle$ an invariant spin- j' multiplet of $(\Pi, \mathcal{H})$, with α, α' any further labels. Then*

$$\langle \alpha' j' m' | T_q^k | \alpha j m \rangle = \langle j m; k q | j' m' \rangle \langle \alpha' j' || T^k || \alpha j \rangle : \quad (12.6)$$

the entire dependence on (m', q, m) is the Clebsch–Gordan coefficient, and one number — the reduced matrix element — carries all the dynamics. (Conventions differ by factors such as $\sqrt{2j'+1}$; ours is fixed by (12.6).) In particular the matrix element vanishes unless

$$m' = m + q \quad \text{and} \quad |j - k| \leq j' \leq j + k.$$

Proof. Lecture 7’s intertwiner argument, at its sharpest. Define (reusing the letter Φ for this intertwiner only — it is not the double cover here) $\Phi: V_j \otimes \mathbb{C}^{2k+1} \rightarrow \mathcal{H}$ by $|j, m\rangle \otimes |k, q\rangle \mapsto T_q^k |\alpha j m\rangle$. It intertwines: by (12.5),

$$\Pi(U) T_q^k |\alpha j m\rangle = (\Pi(U) T_q^k \Pi(U)^{-1}) \Pi(U) |\alpha j m\rangle = \sum_{q', m'} D_{q'q}^k(U) D_{m'm}^j(U) T_{q'}^k |\alpha j m'\rangle,$$

which is Φ applied to $(\Pi_j \otimes \Pi_k)(U)$ of the argument. Let P be the orthogonal projection onto the target multiplet — an intertwiner, since the multiplet and its complement are invariant under the unitary Π . Then $P\Phi$ is an intertwiner from $\Pi_j \otimes \Pi_k$ to $\Pi_{j'}$, and by the Clebsch–Gordan series (12.4) the spin j' occurs in $\Pi_j \otimes \Pi_k$ *at most once*: by Schur, the space of such intertwiners is at most one-dimensional, spanned by the coupled-basis projection $|j, m\rangle \otimes |k, q\rangle \mapsto \langle j m; k q | j' m' \rangle |j', m'\rangle$ (summed over m'). So $P\Phi = \lambda \cdot$ (that map), the scalar λ is the reduced matrix element, and taking components gives (12.6); the selection rules

are the support of the Clebsch–Gordan coefficient. Where Lecture 7’s finite groups allowed multiplicities and several reduced constants, $SU(2)$ ’s multiplicity-free series leaves exactly one. ■

The same conclusion holds for the domain-qualified extension stated after Definition 12.14, provided both displayed multiplets lie in $\mathcal{D}$. Indeed $P_j T_q^k|_{V_j}$ is then an ordinary map between finite-dimensional spaces, and the preceding intertwiner proof applies verbatim.

Proposition 12.17 (Projection theorem; proof on Sheet 12). *Within a single multiplet with $j > 0$, every vector operator is proportional to $\vec{J}$:*

$$\langle \alpha j m' | V_q | \alpha j m \rangle = \frac{\langle \vec{J} \cdot \vec{V} \rangle_{\alpha j}}{j(j+1)} \langle j m' | J_q | j m \rangle,$$

where $\langle \vec{J} \cdot \vec{V} \rangle_{\alpha j}$ is the (constant) expectation of the scalar (rank-0) operator $\vec{J} \cdot \vec{V}$ on the multiplet — constant on it by the $k = 0$ case of Wigner–Eckart.

For a singlet ($j = 0$), the Wigner–Eckart selection rule instead gives $\langle \alpha 0 0 | V_q | \alpha 0 0 \rangle = 0$: every within-multiplet matrix element of a vector operator vanishes. This is a separate statement; the quotient in the projection theorem is not defined at $j = 0$.

12.5 Applications: selection rules and the Zeeman effect

Example 12.18 (Electric dipole selection rules). In the dipole approximation, light couples to an atom through the position operator $\vec{x}$ — a rank-one tensor operator, whose spherical components are proportional to $r Y_{1q}(\hat{x})$. A transition amplitude between orbital states is $\langle n' \ell' m' | x_q | n \ell m \rangle$. The bound-state multiplets used here lie in the domain of x_q and in a common rotation-invariant domain of the kind stated above. Thus this is the domain-qualified finite-multiplet extension, not a claim that x_q is bounded, and Theorem 12.16 clears the table:

$$\Delta m = q \in \{0, \pm 1\}, \quad \ell' \in \{\ell - 1, \ell, \ell + 1\},$$

which polarisation drives which Δm being a routine bookkeeping matter. Parity removes the middle option: under $x \mapsto -x$ the multiplet $\mathcal{Y}_\ell$ carries the sign $(-1)^\ell$ — the fact Lecture 11 packed for this trip — while $\vec{x}$ is odd, so the matrix element acquires $(-1)^{\ell'}(-1)(-1)^\ell$ and vanishes unless $\ell + \ell'$ is odd. Hence the textbook rules

$$\Delta \ell = \pm 1, \quad \Delta m \in \{0, \pm 1\} :$$

the same two-step logic — decompose, then intertwine — that produced Lecture 7’s molecular selection rules, now with integrals under the hood.

Example 12.19 (Zeeman effect and the Landé g -factor). An electron with orbital and spin angular momentum, $\mathcal{H} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^2$ and $\vec{J} = \vec{L} + \vec{S}$ in the sense of (12.3), feels a weak field $\vec{B} = B\hat{z}$ through $H_Z = \mu_B B (L_z + g_s S_z)$ with $g_s = 2$. First-order shifts within a fixed (ℓ, s, j) multiplet with $j > 0$ are diagonal expectations (since $[H_Z, J_z] = 0$, H_Z is diagonal in the $|j, m\rangle$ basis, so degenerate perturbation theory reduces to expectations), and $L_z + 2S_z = J_z + S_z$. Now $\vec{S}$ is a vector operator, so the projection theorem applies:

$$\langle S_z \rangle = \frac{\langle \vec{J} \cdot \vec{S} \rangle}{j(j+1)} m, \quad \vec{J} \cdot \vec{S} = \frac{1}{2}(J^2 + S^2 - L^2) = \frac{1}{2}(j(j+1) + s(s+1) - \ell(\ell+1)),$$

the second identity from squaring $\vec{L} = \vec{J} - \vec{S}$ and the Casimir values of Lecture 10. Hence $\Delta E = \mu_B B g_J m$ with

$$g_J = 1 + \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)} : \quad (12.7)$$

the Landé g -factor, with the anomalous pattern of Zeeman multiplets falling out of one projection (full bookkeeping on Sheet 12). Each level splits into $2j+1$ equally spaced lines — the multiplet made visible, dimension theorem on a photographic plate.

Step back and hear the click. Everything Lecture 7 did for molecules — average, characters, decompose, intertwine, select — this lecture has done for atoms, with the single substitution $\frac{1}{|G|} \sum \rightarrow \int_G$. Part I’s finite machinery was not a toy version of the continuous theory; it *was* the theory, waiting for a better measure. Next, the machinery meets its first genuine obstruction: Lecture 13’s Heisenberg group has no faithful finite-dimensional unitary representations at all — a two-line trace argument — and quantum mechanics will force the course, at last, into infinite dimensions, where the oscillator, Fock space, and Part I’s symmetric group are waiting.

Remark 12.20 (Computer algebra). Notebook 12 integrates over $SU(2)$ twice — by the parametrised class formula (12.1) and by Monte Carlo over random unitaries — and watches the answers agree; verifies character orthonormality to machine precision; rebuilds Clebsch–Gordan coefficients by the ladder algorithm and checks them against `ClebschGordan` and `ThreeJSymbol`; and decomposes $j_1 \otimes j_2$ by numerical character inner products — Part I’s projector code, ported from sums to integrals.

Summary.

- *Haar measure.* Every compact group carries a bi-invariant probability measure (Theorem 12.1); on $SU(2)$ it integrates class functions with weight $\sin^2(\theta/2)$ (Theorem 12.5), whence unitarisability and Maschke (Theorem 12.3).
- *Orthogonality.* The $SU(2)$ characters are orthonormal (Theorem 12.6) and complete (Theorem 12.9).
- *Characters.* $\chi_j(\theta) = \sin((2j+1)\theta/2)/\sin(\theta/2)$ (Theorem 12.8).
- *Clebsch–Gordan.* $j_1 \otimes j_2 = \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} j$, from the product rule for characters (Theorem 12.12, Proposition 12.11).
- *Wigner–Eckart.* A tensor operator’s matrix elements factor into a Clebsch–Gordan coefficient and one reduced matrix element (Theorem 12.16), giving selection rules and the Landé formula (eq. (12.7)).

Tutorial. Exercise Sheet 12.

References

Hall Ch. 4 (compact groups, complete reducibility), Appendix C, Section C.2, Theorem C.6 (Wigner–Eckart), and the appendix on Haar measure; Tung (group integration; Wigner–Eckart); Jones Pt. 9–10; Elliott–Dawber (angular-momentum coupling); Ma (Clebsch–Gordan coefficients); Stein–Shakarchi, Ch. 3, §1, Theorem 1.1, pp. 69–79 (mean-square completeness of Fourier series, used in the character-completeness argument).

Bibliography

The per-lecture *References* sections cite the works below by author and short title; full details are collected here. Chapter and section numbers in the lecture references follow these editions.

- M. A. Armstrong, *Groups and Symmetry*. Springer (UTM), 1988.
- V. Bargmann, On unitary ray representations of continuous groups. *Annals of Mathematics* **59** (1954), 1–46.
- A. O. Barut and R. Rączka, *Theory of Group Representations and Applications*, 2nd ed. World Scientific, 1986.
- R. Berndt, *Representations of Linear Groups: An Introduction Based on Examples from Physics and Number Theory*. Vieweg, 2007. DOI: 10.1007/978-3-8348-9401-4. Theorem 5.2, p. 71 gives the SNAG theorem used for the translation subgroup.
- D. M. Bishop, *Group Theory and Chemistry*. Dover, 1993 (orig. Oxford University Press, 1973).
- O. Bratteli and D. W. Robinson, *Operator Algebras and Quantum Statistical Mechanics 2: Equilibrium States, Models in Quantum Statistical Mechanics*, 2nd ed. (1997), 2nd printing. Springer, 2002. Section 5.2.1, pp. 7–8 gives Fock-space second quantisation, its finite-particle analytic core and closure, and the lifted unitary group.
- G. Costa and G. Fogli, *Symmetries and Group Theory in Particle Physics*. Springer (Lecture Notes in Physics 823), 2012.
- J. B. Conway, *A Course in Functional Analysis*, 1st ed. Springer (GTM 96), 1985. Chapter IX, Theorems 6.4 and 6.6 give the strong/weak bicommutant closure and the multiplication maximal-abelian theorem for sigma-finite measure spaces.
- J. P. Elliott and P. G. Dawber, *Symmetry in Physics*, 2 vols. Macmillan, 1979.
- K. Erdmann and M. J. Wildon, *Introduction to Lie Algebras*. Springer (SUMS), 2006.
- W. Fulton and J. Harris, *Representation Theory: A First Course*. Springer (GTM 129), 1991.
- G. B. Folland, *Harmonic Analysis in Phase Space*. Princeton University Press (Annals of Mathematics Studies 122), 1989. Chapter 1, Sections 2–5, including Theorem 1.50, treats the Heisenberg group, Schrödinger/Weyl representation, and Stone–von Neumann theorem.
- G. B. Folland, *A Course in Abstract Harmonic Analysis*, 2nd ed. Chapman & Hall/CRC, 2016. Section 6.6, Theorem 6.42, pp. 199–201 gives the regular semidirect-product Mackey theorem, including induced irreducibility, exhaustivity, and equivalence.
- R. Gilmore, *Lie Groups, Lie Algebras, and Some of Their Applications*. Wiley, 1974 (repr. Dover, 2005).
- B. C. Hall, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, 2nd ed. Springer (GTM 222), 2015.
- N. Jeevanjee, *An Introduction to Tensors and Group Theory for Physicists*, 2nd ed. Birkhäuser, 2015.
- H. F. Jones, *Groups, Representations and Physics*, 2nd ed. Institute of Physics Publishing,

1998. Chapter 8, Sections 8.3–8.4, pp. 149–164 is the course’s explicit $SU(3)$ further-reading route; Chapter 10, Section 10.4, pp. 208–216 treats the Poincaré little groups; Appendix E, pp. 279–280 derives the covariant on-shell measure from $\theta(p^0)\delta(p^2 - m^2)d^4p$.
- Z.-Q. Ma, *Group Theory for Physicists*. World Scientific, 2007.
 - A. M. L. Messiah and O. W. Greenberg, Symmetrization Postulate and Its Experimental Foundation. *Physical Review* **136** (1964), B248–B267. DOI: <https://doi.org/10.1103/PhysRev.136.B248>.
 - H. Osborn, *Group Theory*. University of Cambridge, Mathematical Tripos Part III lecture notes (available from the DAMTP lecture-notes pages).
 - M. Reed and B. Simon, *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*, rev. ed. Academic Press, 1980. Chapter VIII, Theorems VIII.5, VIII.8, VIII.10, and VIII.13 and Section VIII.5, Example 2 cover spectral calculus, Stone’s theorem, invariant-domain generators, strong commutation, and the bounded-CCR obstruction.
 - P. Ramond, *Group Theory: A Physicist’s Survey*. Cambridge University Press, 2010. Chapter 11, Section 11.1.1, pp. 224–231 treats the Pauli–Lubanski Casimirs and the massless little group; pp. 230–231 distinguish finite helicity from continuous spin and record the two integer/half-odd-integer cover sectors.
 - H. Rauch, A. Zeilinger, G. Badurek, A. Wilfing, W. Bauspiess, and U. Bonse, Verification of Coherent Spinor Rotation of Fermions. *Physics Letters A* **54** (1975), 425–427. DOI: 10.1016/0375-9601(75)90798-7.
 - J. Schwichtenberg, *Physics from Symmetry*, 2nd ed. Springer, 2018.
 - J.-P. Serre, *Linear Representations of Finite Groups*. Springer (GTM 42), 1977.
 - J. Stillwell, *Naive Lie Theory*. Springer (UTM), 2008.
 - E. M. Stein and R. Shakarchi, *Fourier Analysis: An Introduction*. Princeton University Press (Princeton Lectures in Analysis I), 2003. Chapter 3, Section 1, Theorem 1.1, pp. 69–79 gives the mean-square completeness result used in Lecture 12.
 - M. Tinkham, *Group Theory and Quantum Mechanics*. McGraw–Hill, 1964 (repr. Dover, 2003).
 - W.-K. Tung, *Group Theory in Physics*. World Scientific, 1985. Chapter 10, Section 10.4, pp. 191–200 treats induced Poincaré representations, Pauli–Lubanski brackets, and massive and massless little groups.
 - S. Weinberg, *The Quantum Theory of Fields, Vol. I: Foundations*. Cambridge University Press, 1995. Section 5.6 treats causal fields and the relativistic spin–statistics connection, a theorem not derived from a rotation phase in this course.
 - S. A. Werner, R. Colella, A. W. Overhauser, and C. F. Eagen, Observation of the Phase Shift of a Neutron Due to Precession in a Magnetic Field. *Physical Review Letters* **35** (1975), 1053–1055. DOI: 10.1103/PhysRevLett.35.1053.
 - P. Woit, *Quantum Theory, Groups and Representations: An Introduction*. Springer, 2017.